University of Florida, Department of Mathematics

Analysis Qualifying Exam, January 2026, Part 1

Write solutions in a neat and logical fashion, giving complete reasons for all steps. Do three of four.

1. Do two of three.

(i) Give an example of a Lebesgue measurable set $E \subseteq [0,1]$ that is closed, has positive measure, but contains no non-trivial interval or explain why no such example exists.

(ii) Is the function $g : \mathbb{R} \to \mathbb{R}$ defined by $g(x) = \frac{1}{1+x^2}$ the Hardy–Littlewood maximal function of an $\mathcal{L}^1(\mathbb{R})$ function f?

(iii) Does there exist a measure space $(X, \mathcal{M}, \mu)$ and a nested decreasing sequence $(E_n)_{n=1}^{\infty}$ of measurable sets such that

$$\lim_{n\to\infty} \mu(E_n) \neq \mu\left(\bigcap_{n=1}^{\infty} E_n\right)?$$

2. Suppose $f : [0,1] \times [0,1] \to \mathbb{R}$. Show, if (a)–(c), then $F : [0,1] \to \mathbb{R}$ defined by

$$F(x) = \int_{[0,1]} f(x,y)\, d\lambda(y)$$

is continuous. Here λ denotes Lebesgue measure on $\mathbb{R}$.

(a) for each $x \in [0,1]$ the function $g_x(y) = f(x,y)$ is continuous;

(b) for each $y \in [0,1]$ the function $h_y(x) = f(x,y)$ is continuous; and

(c) f is bounded,

3. Let $(X, \mathcal{M})$ denote a measurable space and let $P(X)$ denote the power set of X. Prove, if $\mathcal{M} \neq P(X)$ and $\{x\} \in \mathcal{M}$ for each $x \in X$, then there exists an $A \notin \mathcal{M} \otimes \mathcal{M}$ such that $[A]_x, [A]^y \in \mathcal{M}$ for all $x \in X$. Suggestion: First show $\delta : X \to X \times X$ defined by $\delta(x) = (x,x)$ is measurable from $(X, \mathcal{M})$ to $(X \times X, \mathcal{M} \otimes \mathcal{M})$.

4. Suppose $(X, \mathcal{M}, \mu)$ is a measure space and $f, g : X \to [0, \infty)$ are measurable. Show

(a) $\mu_f : \mathcal{M} \to [0, \infty]$ defined by $\mu_f(A) = \int \chi_A f\, d\mu = \int_A f\, d\mu$ is a measure; and

(b) $\int g\, d\mu_f = \int g f\, d\mu$.